Amrit SS

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EDUCATION

Vellore Institute of Technology (VIT),Chennai 2022-2026
Bachelor of Technology- Mechatronics and automation Chennai,Tamilnadu

WORK EXPERIENCE

QUALITY CONTROL MANAGER **June 2024 –July 2024**
Schneider Electric

- **Improved Defect Analysis**: Identified and categorized manufacturing defects in the thermal subassembly station, leading to a clearer understanding of recurring issues. Decreased MCB defects by 3 percent in my sector and AGV interaction with coding.
- **Pareto Analysis Implementation**: Conducted data-driven analysis to prioritize major defect causes, enabling focused corrective actions. Identifying that 80% of defects were caused by 20% of issues (80/20 rule).

ELECTRICAL AND ELECTRONICS ENGINEER
Team Movis(VIT CHENNAI) **March. 2024 – Present**

- **Sensor & Actuator Integration**: Integrated various sensors for terrain mapping and actuators for precise rover movement.
- **ROS Integration & Robotics Control –** Implemented sensor fusion, path planning, and autonomous decision-making using ROS.
- **Communication Systems:** Worked on data transmission for reliable control and feedback in rover operations.
- **Power Management & Telemetry:** Optimized power consumption and developed remote monitoring systems for real-time diagnostics**.**

PROJECTS

NASA HERC (Human Exploration Rover Challenge 2026)

- **Chassis Design and Material Selection:** Analyzed and selected lightweight materials like aluminum and carbon fiber for optimal strength and weight efficiency**.**
- **Powertrain Development:** Designed a powertrain with the right balance of speed, torque, and energy efficiency for rugged terrain navigation**.**
- **Suspension System Design**
- **Telemetry and Communication:** Implemented telemetry systems for real-time data collection and remote-control during rover testing and operation

ISRO Drone Challenge – Autonomous Topology Mapping Drone

- **Drone Design and Aerodynamics:** Designed a lightweight, durable drone frame optimized for stability and efficient flight in varying conditions.
- **Mapping and Topology Algorithms.**
- **Flight Control System:** Programmed the flight control system to handle autonomous navigation, Tracking return path, and smooth landings in diverse environments**.**

FLIGHT SIMULATOR CONTROL INPUT LOGGER

- Built a system to log pilot inputs (yoke, rudder, throttle) with millisecond precision.
- Supported training analysis and simulator performance evaluation.

CATERPILLAR HACKATHON RUNNER UP

- Selected among top 3000 teams for the finals of the Caterpillar Hackathon for developing a real-time monocular depth estimation system on edge AI devices and used CAD design for the complete idea of implementation.

RFID-Based Automatic Pill Dispenser

- Developed an RFID authentication system for secure medication dispensing.

CHLOROGAUGE- (IN SITU CHLORPHYLL MONITORING PRODUCT)- TRL 4

- Developed an **agricultural chlorophyll gauge** validated at **Technology Readiness Level (TRL) 4** through functional laboratory testing.
- Implemented optical sensing techniques for non-destructive estimation of **leaf chlorophyll content and crop health assessment.**

Face Recognition Attendance System (Python & OpenCV)

- Implemented real-time face detection and recognition using OpenCV.
- Developed a database system for storing and managing attendance logs.
- Optimized accuracy and recognition speed for practical deployment.

ROS BASED AUTONOMOUS WAREHOUSE AMR

- Integrated LiDAR and ultrasonic sensors for environment perception and obstacle detection.
- Developed URDF model and simulated navigation stack using ROS2 and Gazebo

- Fabrication of a custom Scissor mechanism with linear actuator

TurtleBot simulations with Gazebo and Rviz

- Performed multiple ROS2 simulations using TurtleBot3 in Gazebo.
- Implemented SLAM for environment mapping and autonomous navigation.
- Configured Nav2 stack for path planning and dynamic obstacle avoidance.
- Experimented with different world setups to simulate indoor navigation tasks

LEADERSHIP

Head Organizers
Voyage 2024 (VIT CHENNAI-ENTREPRENEUR EXPO)
Stage committee (VIBRANCE 2025 – VIT CHENNAI)

SKILLS

SOFTARE TOOLS : ROS2 (Robotic operating system), C/C++, JAVA BASICS, SOLIDWORKS, SCADA,PLC, MATLAB SIMULINK, SALESFORCE, ROBOSTUDIO, GAZEBO
LANGUAGES: English, Tamil, Saurashtra, **German (A2 PROFICIENCY)**
French and Hindi: Limited working proficiency.